



MGM's College of Engineering, Nanded

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AI-Powered Podcast Analysis and Live Captioning System

Name of Student: Aditi Chintawar

Name of Guide: Mr. S.N.Neralkar

Academic Year : 2025-26

Introduction: Podcasts are a rich source of information, but extracting meaningful insights and making content accessible in real time can be challenging. The AI-Powered Podcast Analysis and Live Captioning System is a web application that processes podcast audio and converts it into actionable insights. It performs speech-to-text transcription, topic segmentation, summarization, sentiment analysis, keyword extraction, and provides interactive visual analytics. It also features live captioning synchronized with audio playback and an AI chatbot to answer questions about the podcast content.

System Architecture



• **Upload Module:** User uploads podcast audio file.



• **Speech-to-Text:** Audio is converted to text using Whisper AI.



• **NLP Processing:** Topic segmentation, summarization, sentiment & keyword extraction using NLP models.



• **Analytics Dashboard:** Visual insights through charts, graphs, and word cloud.



• **Chatbot:** AI assistant to answer user queries based on podcast content.



• **Export Module:** Export results to CSV/PDF for future reference.

Methodology

1. User uploads the podcast audio.
2. Speech-to-text conversion using Whisper AI.
3. Topic segmentation using embeddings.
4. Summarization, sentiment analysis & keyword extraction.
5. Dashboard visualization + AI chatbot + Live captions.

AI Chatbot Features

The system has an AI-powered chatbot for podcast-related queries. Users can ask questions in natural language, and it provides instant, accurate responses based on the analyzed podcast content.

Conclusion: The system leverages AI and NLP techniques to automate podcast analysis and provide real-time captions, insights, and interactive assistance. It helps users save time, improve accessibility, and extract valuable knowledge from audio content efficiently.

Mapping of PO's

- **PO2-** Problem Analysis
- **PO2-** Modern Usage
- **PO3-** Design/Development of Solutions
- **PO5-** Modern Tool Usage
- **PO12-** Life-long Learning

Submitted By:

Aditi Chintawar

BTech CSE B

MGM's College of Engineering,
Nanded

Academic Year: 2025-26